



Maastricht University

Department of Data Science and Knowledge Engineering

MATSE: Operations Research

Lecture 6. Intro Stochastic Models: Flipping Coins

[based on notes by: Frank Thuijsman and Steven Chaplick]

Coin Flipping

Suppose you have a coin with two sides H and T .

Coin Flipping

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- Consider a sequence $HHT \cdots HTHT$ of coin flips.

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- What is the **expected** number of coin flips until **first** H ?

Coin Flipping

Suppose you have a coin with two sides H and T .

- Consider a sequence $HHT \cdots HTHT$ of coin flips.
- What is the **expected** number of coin flips until **first** H ?
(i.e., on average over all possible sequences, when does the first H occur?)

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

x	1	2	3	4	5	...	k	
$P(X=x)$								

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

x	1	2	3	4	5	...	k	
$P(X=x)$	$\frac{1}{2}$							

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

x	1	2	3	4	5	...	k	
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$						

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

x	1	2	3	4	5	...	k	
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$					

Coin Flipping

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Let X be the number of coin flips until we see H .

x	1	2	3	4	5	...	k	
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$			

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Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

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$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$		$\frac{1}{2^k}$	

$$\left. \vphantom{\begin{matrix} x \\ P(X=x) \end{matrix}} \right\} \mathbb{E}(X) = ?$$

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

$$\left. \begin{array}{c|c|c|c|c|c|c|c|c} x & 1 & 2 & 3 & 4 & 5 & \dots & k & \\ \hline P(X=x) & \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{16} & \frac{1}{32} & & \frac{1}{2^k} & \end{array} \right\} \mathbb{E}(X) = \sum_{i=1}^{\infty} i \frac{1}{2^i}$$

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

$$\left. \begin{array}{c|c|c|c|c|c|c|c|c} x & 1 & 2 & 3 & 4 & 5 & \dots & k & \\ \hline P(X=x) & \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{16} & \frac{1}{32} & & \frac{1}{2^k} & \end{array} \right\} \mathbb{E}(X) = \sum_{i=1}^{\infty} i \frac{1}{2^i}$$

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Let X be the number of coin flips until we see H .

$$\left. \begin{array}{c|c|c|c|c|c|c|c|c} x & 1 & 2 & 3 & 4 & 5 & \dots & k & \\ \hline P(X=x) & \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{16} & \frac{1}{32} & & \frac{1}{2^k} & \end{array} \right\} \mathbb{E}(X) = \sum_{i=1}^{\infty} i \frac{1}{2^i}$$

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} i \frac{1}{2^i} = 1\frac{1}{2} + 2\frac{1}{4} + 3\frac{1}{8} + 4\frac{1}{16} + \dots$$

Coin Flipping

- What is the **expected** number of coin **flips** until first H ?

Let X be the number of coin flips until we see H .

$$\left. \begin{array}{c|c|c|c|c|c|c|c|c} x & 1 & 2 & 3 & 4 & 5 & \dots & k & \\ \hline P(X=x) & \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{16} & \frac{1}{32} & & \frac{1}{2^k} & \end{array} \right\} \mathbb{E}(X) = \sum_{i=1}^{\infty} i \frac{1}{2^i}$$

$$\begin{aligned} \mathbb{E}(X) &= \sum_{i=1}^{\infty} i \frac{1}{2^i} = 1 \frac{1}{2} + 2 \frac{1}{4} + 3 \frac{1}{8} + 4 \frac{1}{16} + \dots \\ &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\ &\quad + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\ &\quad \quad + \frac{1}{8} + \frac{1}{16} + \dots \\ &\quad \quad \quad + \frac{1}{16} + \dots \end{aligned}$$

Coin Flipping (Computations)

$$\begin{aligned}\sum_{i=1}^{\infty} i \frac{1}{2^i} &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\ &\quad + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\ &\quad \quad + \frac{1}{8} + \frac{1}{16} + \dots \\ &\quad \quad \quad + \frac{1}{16} + \dots\end{aligned}$$

Coin Flipping (Computations)

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 \sum_{i=1}^{\infty} i \frac{1}{2^i} &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad \quad + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad \quad \quad + \frac{1}{16} + \dots
 \end{aligned}$$

$$\sum_{i=1}^{\infty} \frac{1}{2^i}$$

Coin Flipping (Computations)

$$\begin{aligned}
 \sum_{i=1}^{\infty} i \frac{1}{2^i} &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
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 &\quad \quad \quad + \frac{1}{16} + \dots
 \end{aligned}$$

$$\begin{aligned}
 &\sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &\frac{1}{2} \sum_{i=1}^{\infty} \frac{1}{2^i}
 \end{aligned}$$

Coin Flipping (Computations)

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 \sum_{i=1}^{\infty} i \frac{1}{2^i} &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
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 &\quad \quad + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad \quad \quad + \frac{1}{16} + \dots
 \end{aligned}$$

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 \end{aligned}$$

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 &\quad \quad \quad + \frac{1}{16} + \dots
 \end{aligned}$$

$$\begin{aligned}
 &\sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &\frac{1}{2} \sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &\frac{1}{2^2} \sum_{i=1}^{\infty} \frac{1}{2^i} \\
 &= \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \left(\frac{1}{2^i} \sum_{j=1}^{\infty} \frac{1}{2^j} \right)
 \end{aligned}$$

Coin Flipping (Computations)

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 \sum_{i=1}^{\infty} i \frac{1}{2^i} &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad + \frac{1}{8} + \frac{1}{16} + \dots \\
 &\quad + \frac{1}{16} + \dots
 \end{aligned}$$

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 &\frac{1}{2} \sum_{i=1}^{\infty} \frac{1}{2^i} \\
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 &= \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \left(\frac{1}{2^i} \sum_{j=1}^{\infty} \frac{1}{2^j} \right)
 \end{aligned}$$

Coin Flipping (Computations)

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\sum_{j=1}^{\infty} \frac{1}{2^j} \right) =$$

Coin Flipping (Computations)

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Geometric Serie: $\sum_{i=0}^{\infty} q^i = \frac{1}{1-q} \quad q \in (0, 1)$

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$$\sum_{i=1}^{\infty} \frac{1}{2^i} =$$

Coin Flipping (Computations)

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\sum_{j=1}^{\infty} \frac{1}{2^j} \right) =$$

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$$\sum_{i=1}^{\infty} \frac{1}{2^i} = \sum_{i=0}^{\infty} \frac{1}{2^i} - \frac{1}{2^0} =$$

Coin Flipping (Computations)

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\sum_{j=1}^{\infty} \frac{1}{2^j} \right) =$$

Geometric Serie: $\sum_{i=0}^{\infty} q^i = \frac{1}{1-q} \quad q \in (0, 1)$

$$\sum_{i=1}^{\infty} \frac{1}{2^i} = \sum_{i=0}^{\infty} \frac{1}{2^i} - \frac{1}{2^0} = \frac{1}{1 - \frac{1}{2}} - 1 = 1$$

Coin Flipping (Computations)

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\sum_{j=1}^{\infty} \frac{1}{2^j} \right) = \boxed{1} + \sum_{i=1}^{\infty} \frac{1}{2^i} \cdot \boxed{1} = 1 + \boxed{1} = 2$$

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$$\sum_{i=1}^{\infty} \frac{1}{2^i} = \sum_{i=0}^{\infty} \frac{1}{2^i} - \frac{1}{2^0} = \frac{1}{1 - \frac{1}{2}} - 1 = \boxed{1}$$

Coin Flipping (Computations)

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- What is the expected number of coin flips until **first** H ?
2!

Coin Flipping (Computations)

$$\mathbb{E}(X) = \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{i=1}^{\infty} \frac{1}{2^i} \left(\sum_{j=1}^{\infty} \frac{1}{2^j} \right) = \boxed{1} + \sum_{i=1}^{\infty} \frac{1}{2^i} \cdot \boxed{1} = 1 + \boxed{1} = 2$$

Geometric Serie: $\sum_{i=0}^{\infty} q^i = \frac{1}{1-q} \quad q \in (0, 1)$

$$\sum_{i=1}^{\infty} \frac{1}{2^i} = \sum_{i=0}^{\infty} \frac{1}{2^i} - \frac{1}{2^0} = \frac{1}{1 - \frac{1}{2}} - 1 = \boxed{1}$$

- What is the expected number of coin flips until **first** H ?
2!
- 2 is also the inverse of $\frac{1}{2}$ which is the probability of flipping H...

Coin Flipping (cont)

Seems complex ...

Let X be the number of coin flips until we see HT .

Take a Guess!

Coin Flipping (cont)

Seems complex ...

Let X be the number of coin flips until we see HT .

x	1	2	3	4	5	...
$P(X=x)$	0	$\frac{1}{4}$				
		HT				

Coin Flipping (cont)

Seems complex ...

Let X be the number of coin flips until we see HT .

x	1	2	3	4	5	...
$P(X=x)$	0	$\frac{1}{4}$	$\frac{2}{8}$			
		HT	zHT			

Coin Flipping (cont)

Seems complex ...

Let X be the number of coin flips until we see HT .

x	1	2	3	4	5	...
$P(X=x)$	0	$\frac{1}{4}$	$\frac{2}{8}$	$\frac{3}{16}$		
		HT	zHT	$zzHT$ $\neg HTHT$		

Coin Flipping (cont)

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Let X be the number of coin flips until we see HT .

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$P(X=x)$	0	$\frac{1}{4}$	$\frac{2}{8}$	$\frac{3}{16}$	$\frac{4}{32}$	
		HT	zHT	$zzHT$ $\neg HTHT$		

$$\mathbb{E}(X) = \sum_x x \cdot P(X = x)$$

Coin Flipping (cont)

Seems complex ...

Let X be the number of coin flips until we see HT .

x	1	2	3	4	5	...	$\mathbb{E}(X) = \sum_x x \cdot P(X = x)$
$P(X=x)$	0	$\frac{1}{4}$	$\frac{2}{8}$	$\frac{3}{16}$	$\frac{4}{32}$		There has to be a better way!
		HT	zHT	$zzHT$ $\neg HTHT$			

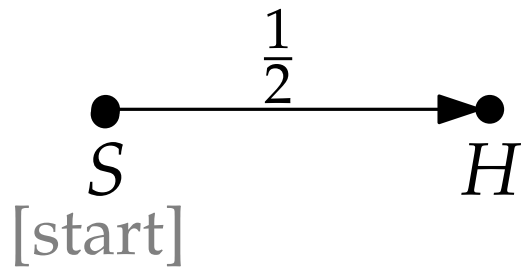
Coin Flipping (cont)

There is a better way!

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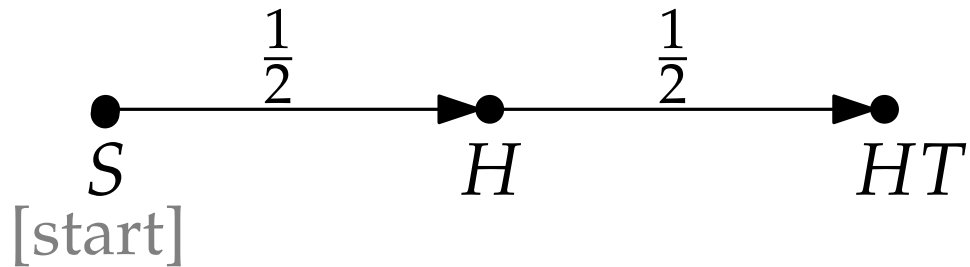
Coin Flipping (cont)

There is a better way!



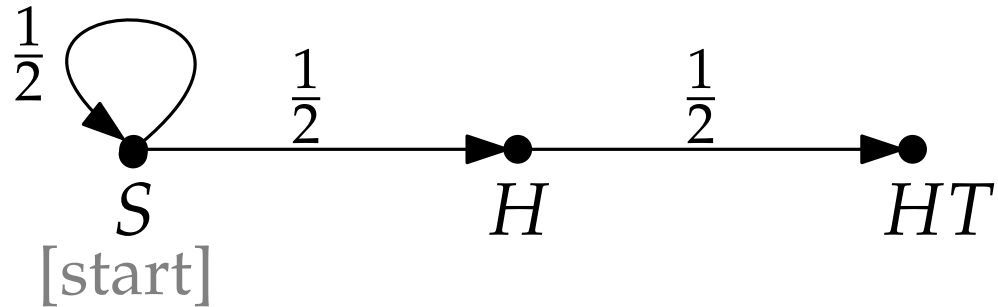
Coin Flipping (cont)

There is a better way!



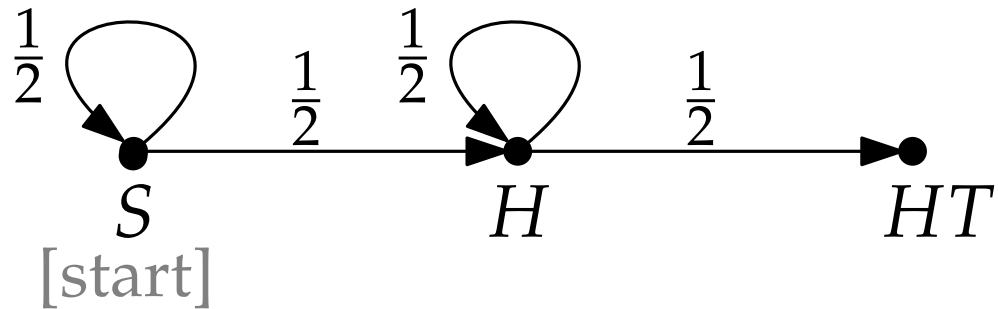
Coin Flipping (cont)

There is a better way!



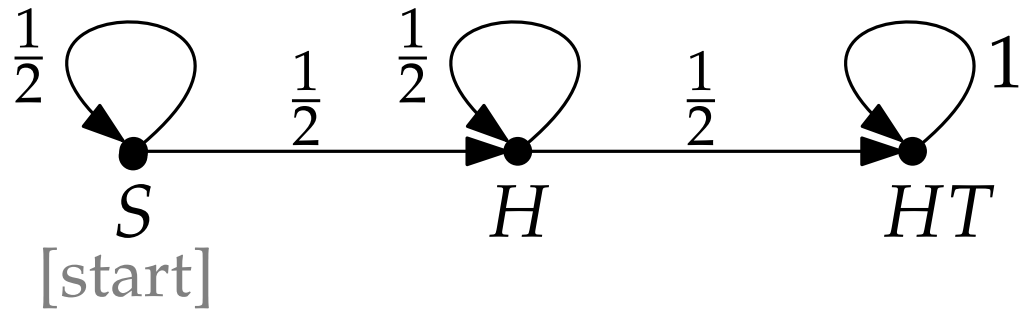
Coin Flipping (cont)

There is a better way!



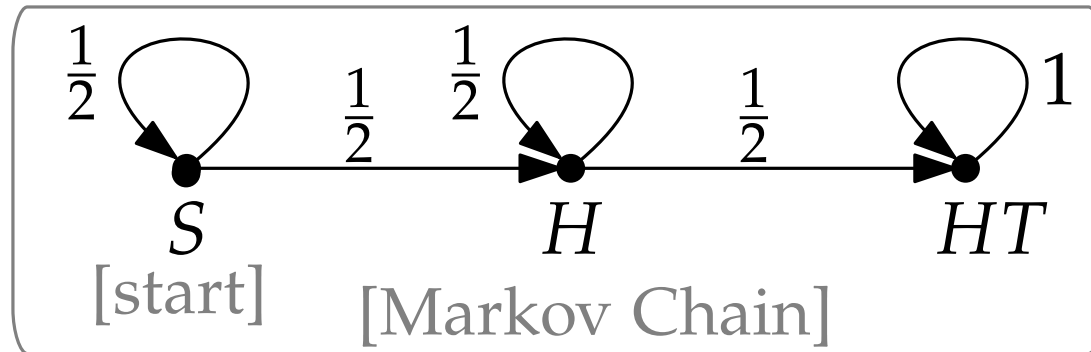
Coin Flipping (cont)

There is a better way!



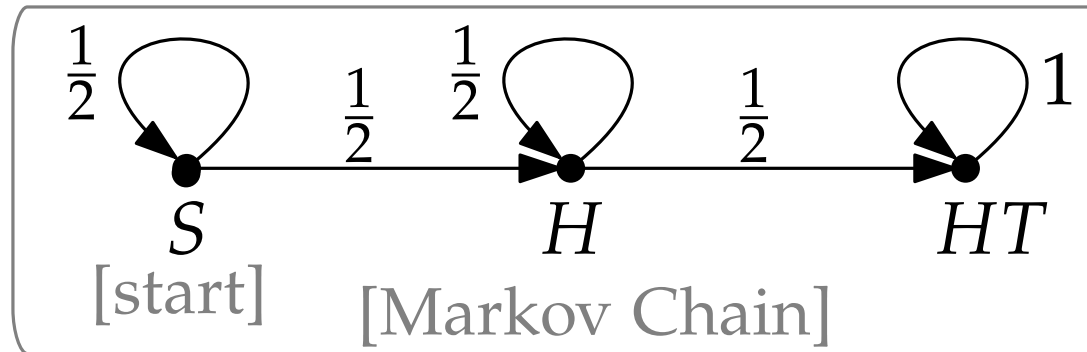
Coin Flipping (cont)

There is a better way!



Coin Flipping (cont)

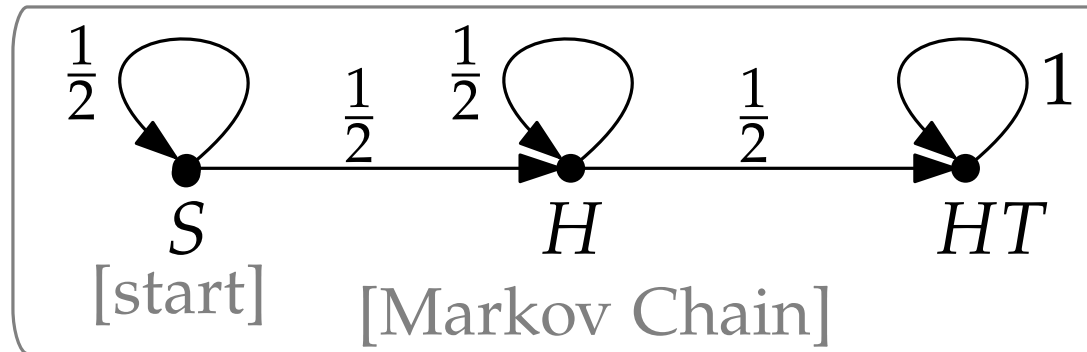
There is a better way!



Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

Coin Flipping (cont)

There is a better way!

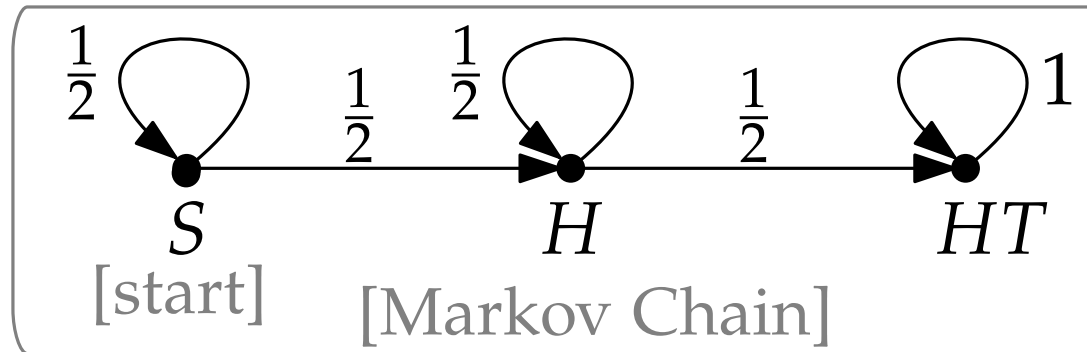


Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

Coin Flipping (cont)

There is a better way!



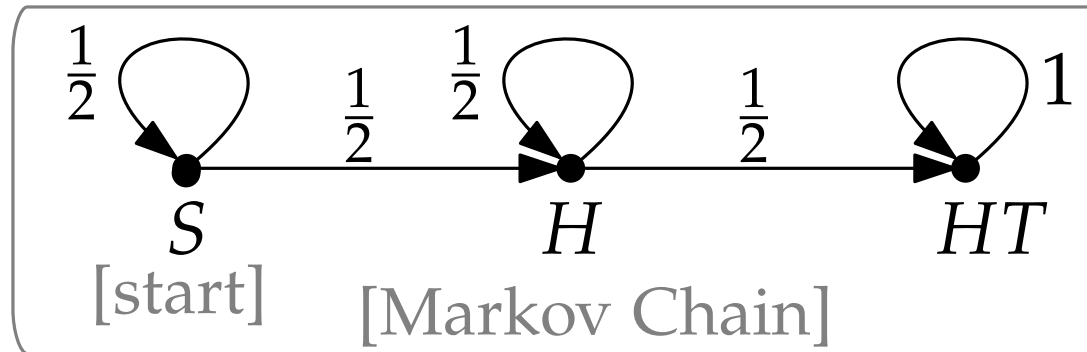
Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT}$$

Coin Flipping (cont)

There is a better way!



Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

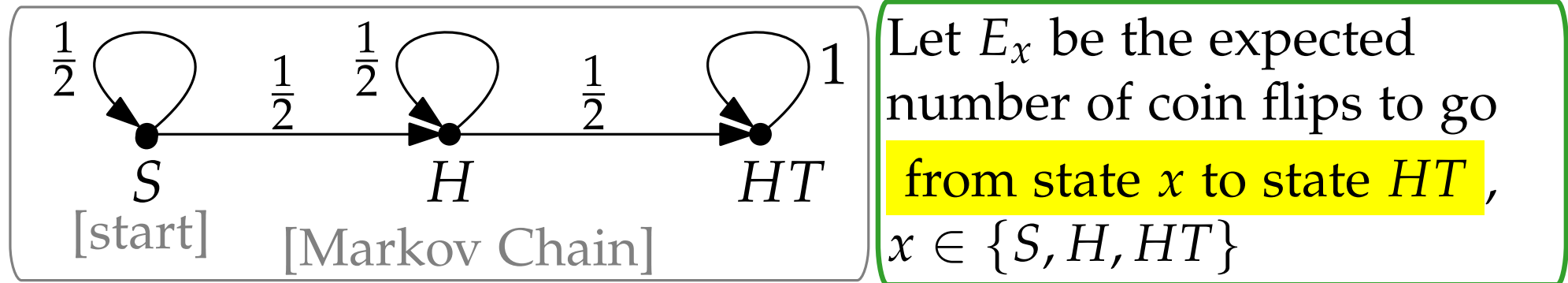
$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT}$$

$$E_{HT} = 0 \text{ [already done!]}$$

Coin Flipping (cont)

There is a better way!



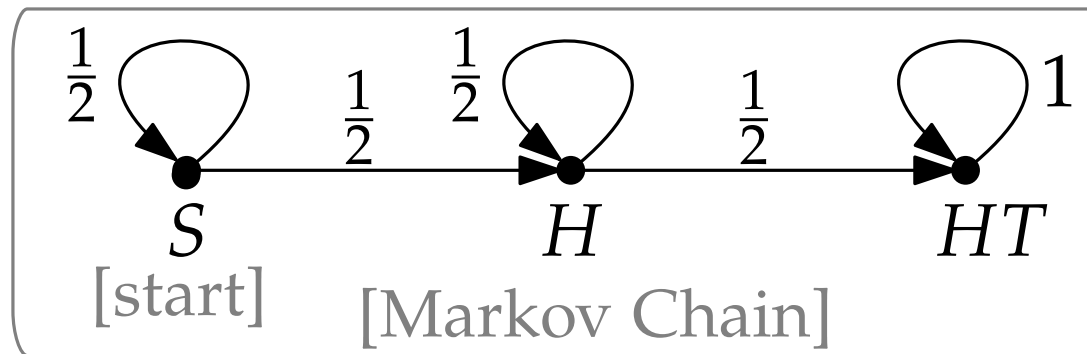
$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT} \rightsquigarrow E_H = 2 \text{ [re-arranging]}$$

$$E_{HT} = 0 \text{ [already done!]}$$

Coin Flipping (cont)

There is a better way!



Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

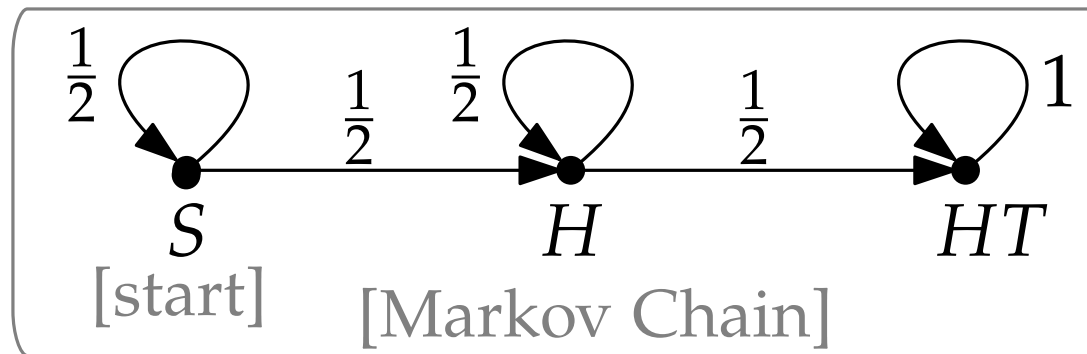
$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \quad \rightsquigarrow \quad E_S = 2 + \frac{1}{2}E_S$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT} \quad \rightsquigarrow \quad E_H = 2 \text{ [re-arranging]}$$

$$E_{HT} = 0 \text{ [already done!]}$$

Coin Flipping (cont)

There is a better way!



Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

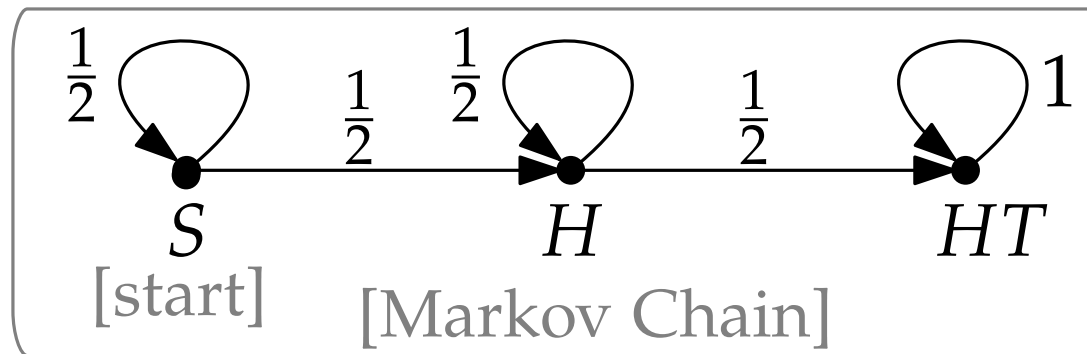
$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \quad \rightsquigarrow \quad E_S = 2 + \frac{1}{2}E_S \quad \rightsquigarrow \quad E_S = 4$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT} \quad \rightsquigarrow \quad E_H = 2 \text{ [re-arranging]}$$

$$E_{HT} = 0 \text{ [already done!]}$$

Coin Flipping (cont)

There is a better way!



Let E_x be the expected number of coin flips to go from state x to state HT , $x \in \{S, H, HT\}$

$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \rightsquigarrow E_S = 2 + \frac{1}{2}E_S \rightsquigarrow E_S = 4$$

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT} \rightsquigarrow E_H = 2 \text{ [re-arranging]}$$

$$E_{HT} = 0 \text{ [already done!]}$$

- 4 is also the inverse of $\frac{1}{4}$ which is the probability of flipping HT...

Coin Flipping (cont)

What is the expected number of coin flips until we see HH .

Take a guess!

Coin Flipping (cont)

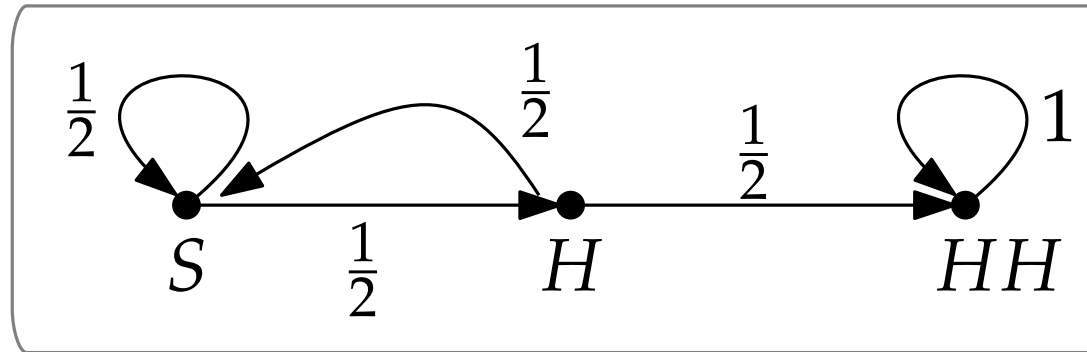
What is the expected number of coin flips until we see HH .

Take a guess!

Can you draw the Markov chain?

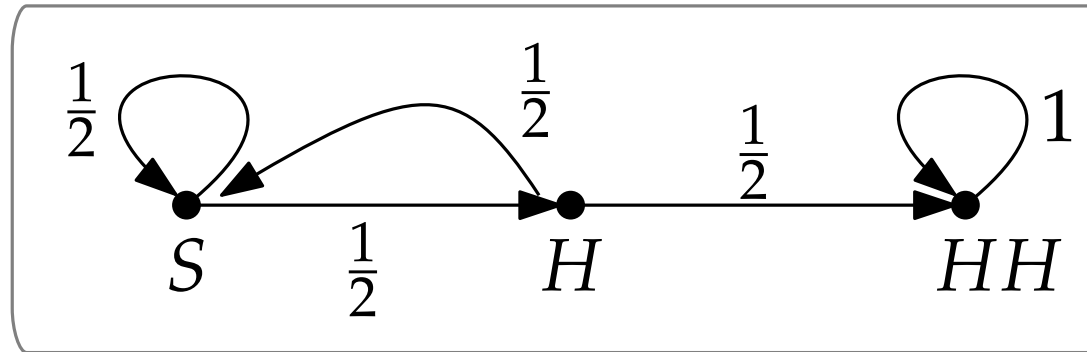
Coin Flipping (cont)

What is the expected number of coin flips until we see HH .



Coin Flipping (cont)

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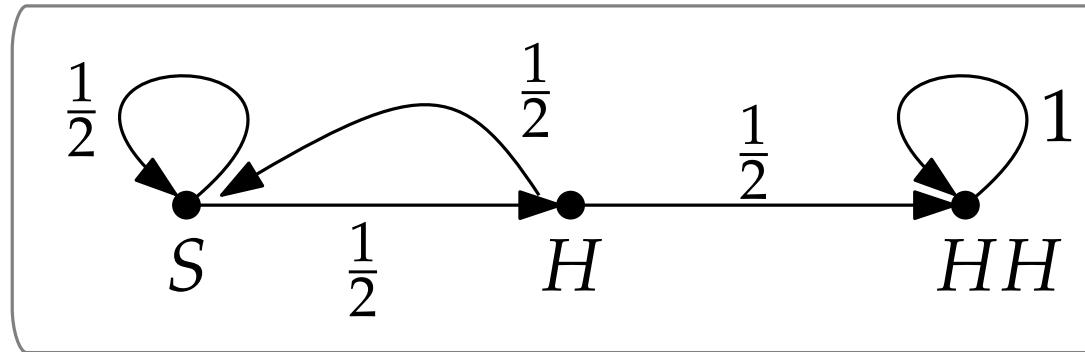
$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

$$E_H = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HH}$$

$$E_{HT} = 0 \text{ [already done!]}$$

Coin Flipping (cont)

What is the expected number of coin flips until we see HH .



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$$E_S = 6!$$

- 6 is not the inverse of $\frac{1}{4}$ which is the probability of flipping HH ...

Coin Flipping: systematically

Let X be the number of coin flips until we see HTH .

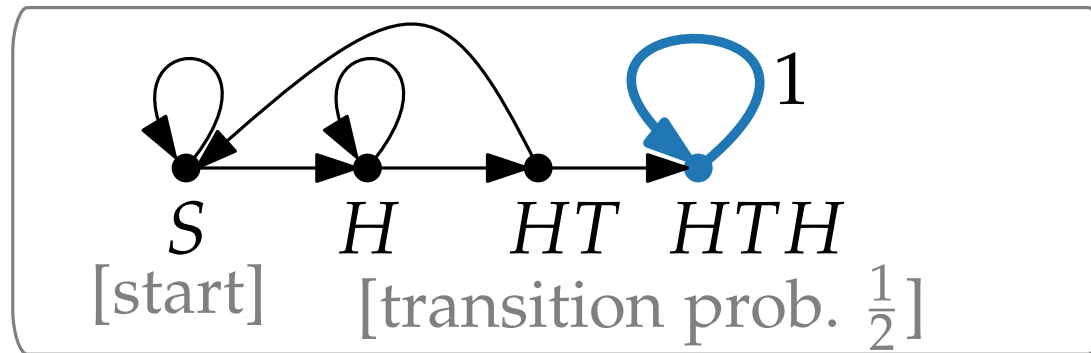
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Try Yourself!

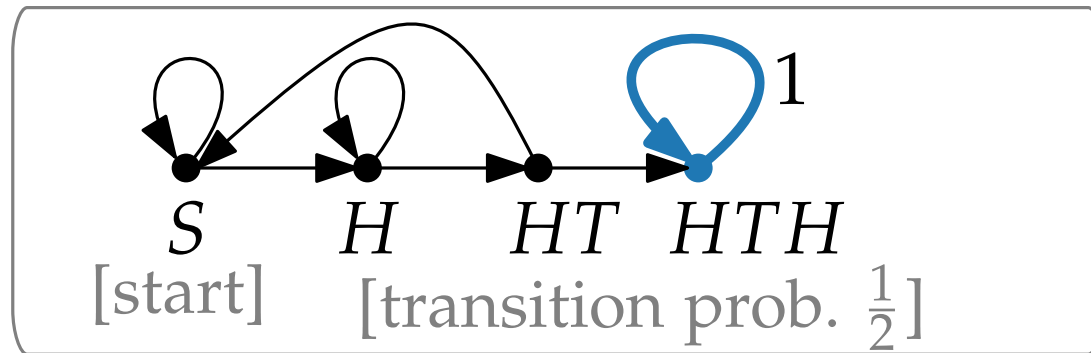
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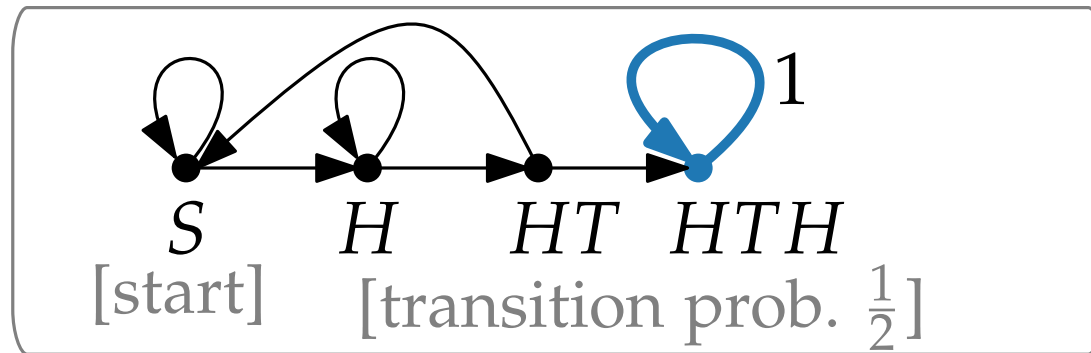
$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT}$$

$$E_{HT} = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HTH}$$

$$E_{HTH} = 0 \text{ [already done!]}$$

Coin Flipping: systematically

Let X be the number of coin flips until we see HTH .



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$$E_{HT} = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HTH}$$

$$E_{HTH} = 0 \text{ [already done!]}$$

$$E_S = 10!$$

- 10 is not the inverse of $\frac{1}{8}$ which is the probability of flipping HH...

Abracadabra Theorem

We actually measured two different things...

Abacadabra Theorem

We actually measured two different things...

- E_x (i.e., the system of equations) measure the number of coin flips before the we see a sequence for the *first* time.
- The inverse of the probability measures the number of coin flips before the process returns *again* to the state

Abracadabra Theorem

We actually measured two different things...

- E_x (i.e., the system of equations) measure the number of coin flips before the we see a sequence for the *first* time.

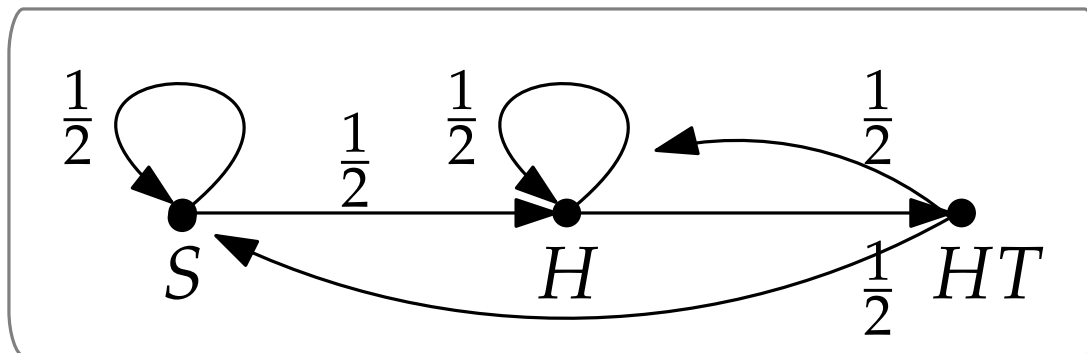
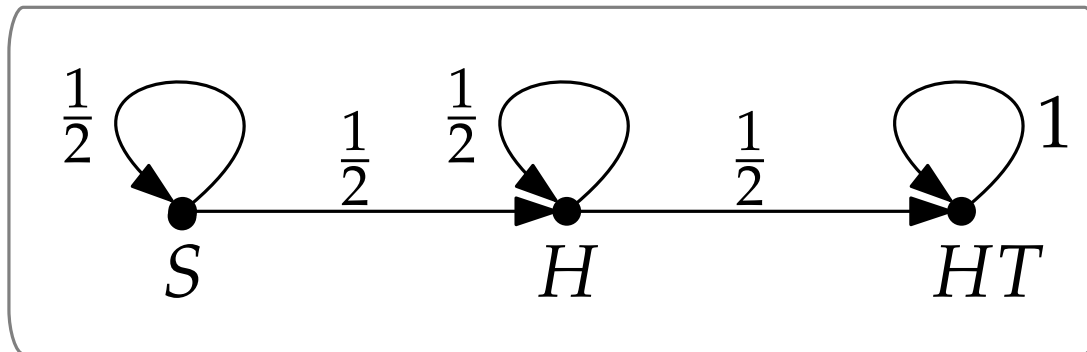
⇒ **First Passage Time**

- The inverse of the probability measures the number of coin flips before the process returns *again* to the state

⇒ **Recurrence Time**

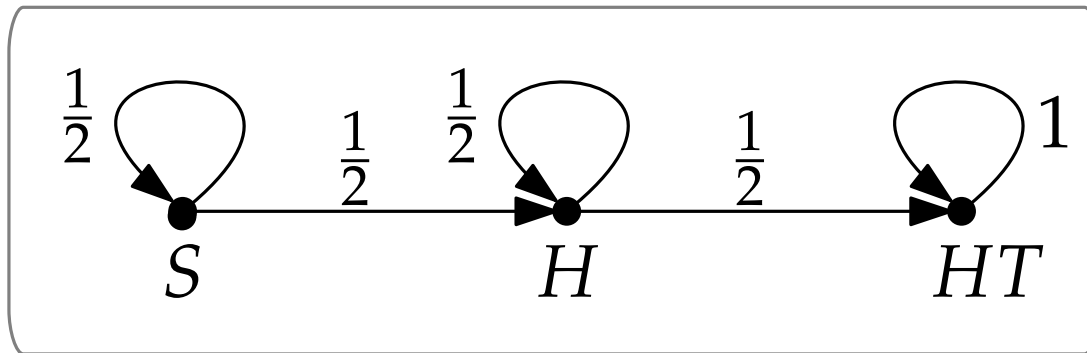
Abracadabra Theorem

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Abracadabra Theorem

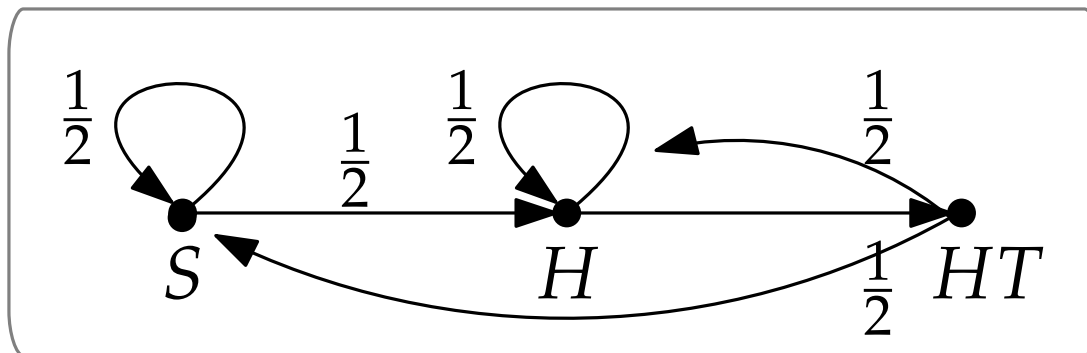
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First Passage Time

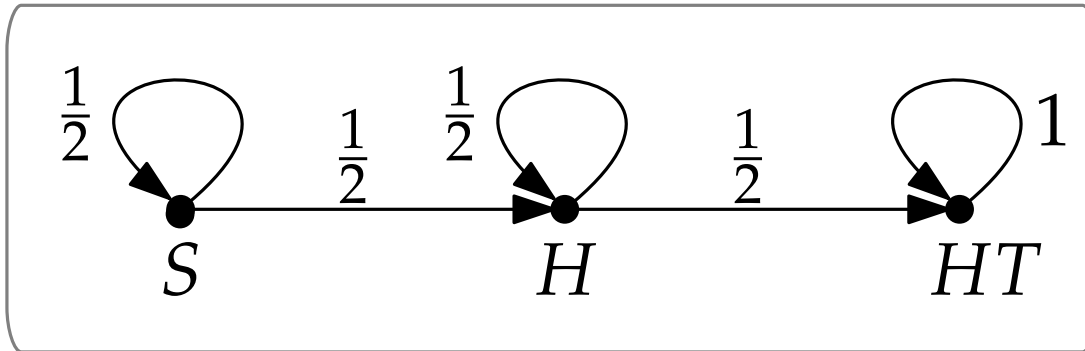
$$E_x = 1 + \sum_j P_{xj} E_j$$

[same as before]



Abacadabra Theorem

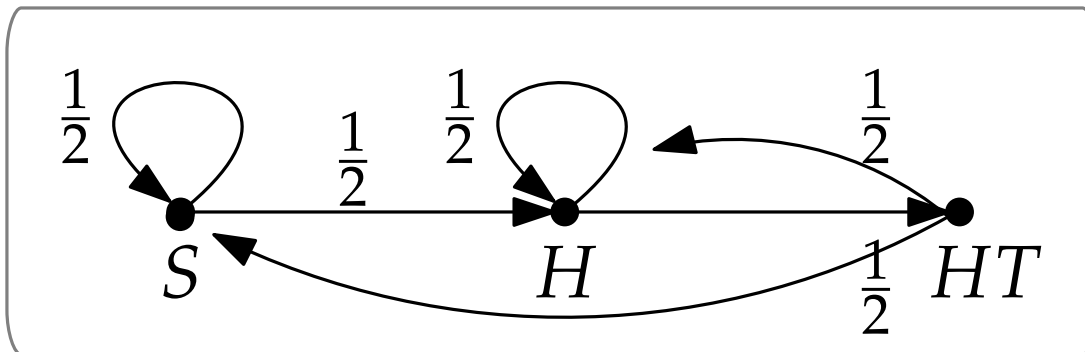
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First Passage Time

$$E_x = 1 + \sum_j P_{xj} E_j$$

[same as before]



Recurrence Time

$$R_x = 1 + \sum_j P_{xj} E_j$$

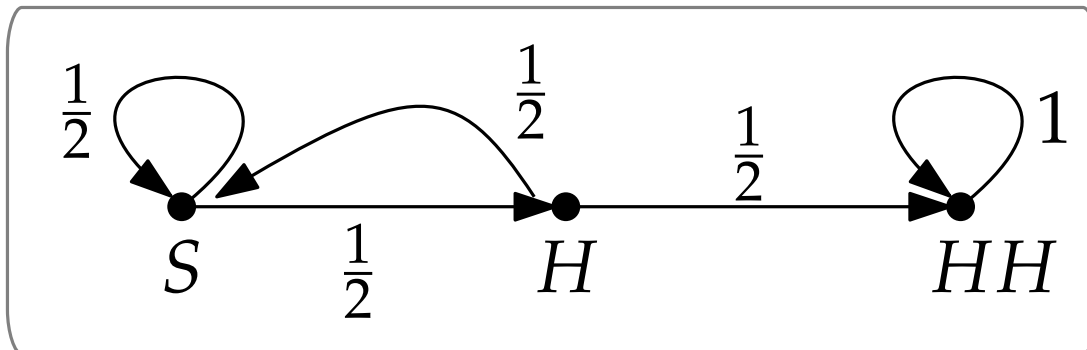
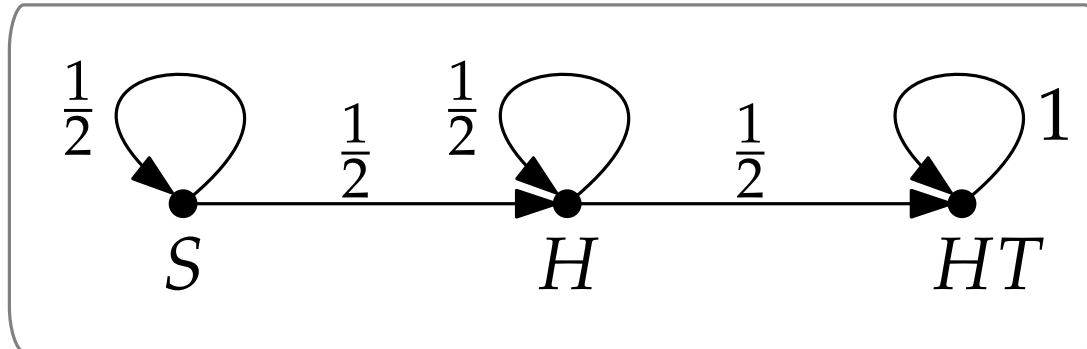
[different chain]

Recurrence time

Is there a faster way?

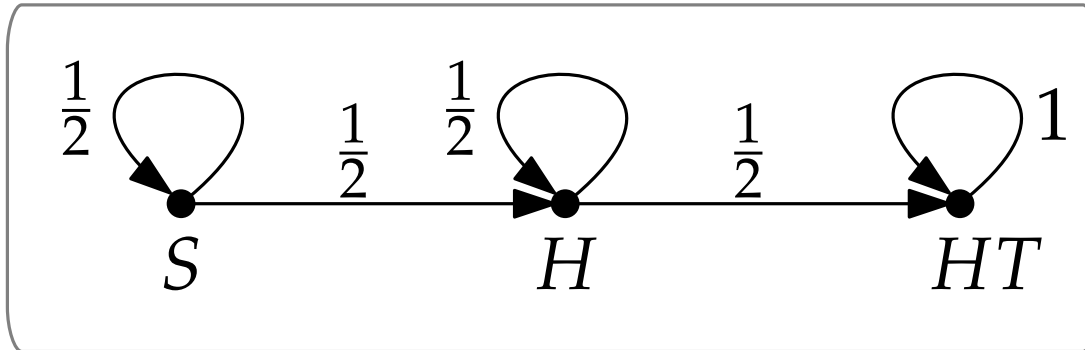
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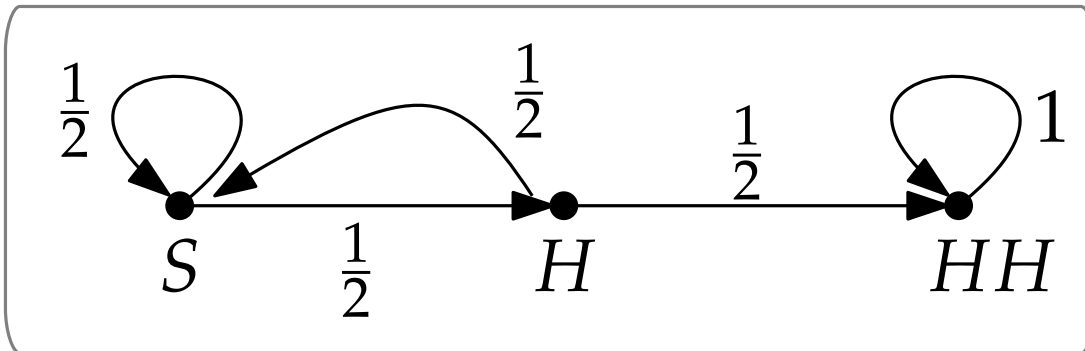


Recurrence time

Is there a faster way?



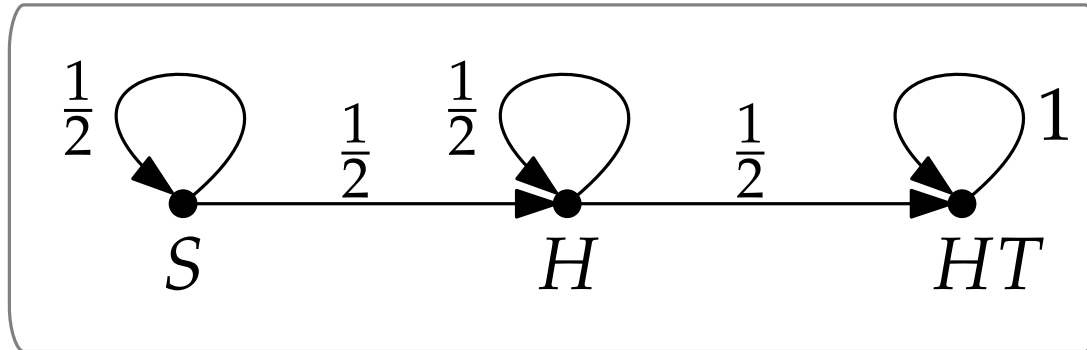
$S \longrightarrow HT$



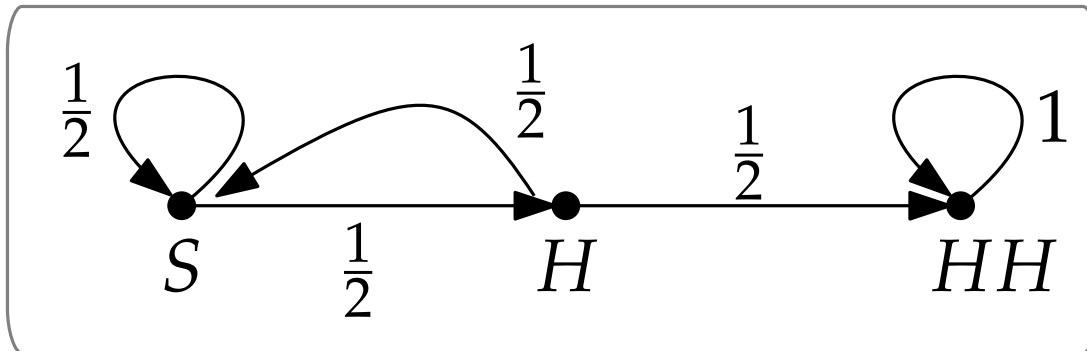
$S \longrightarrow H \longrightarrow HH$

Recurrence time

Is there a faster way?



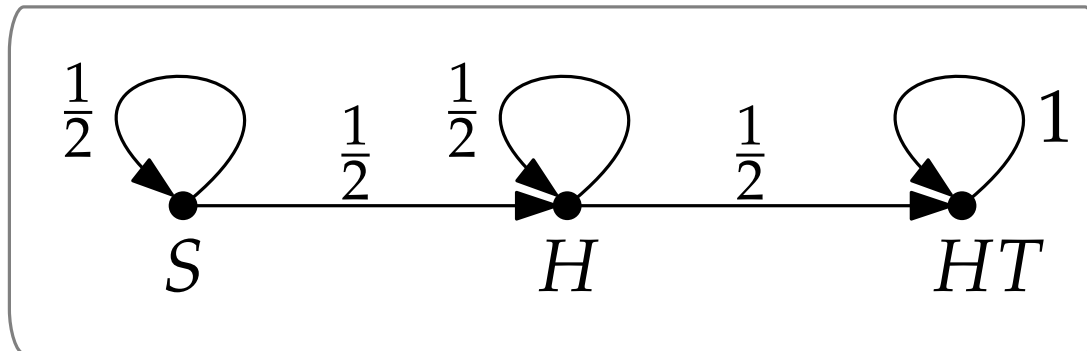
$$S \xrightarrow[4]{} HT$$



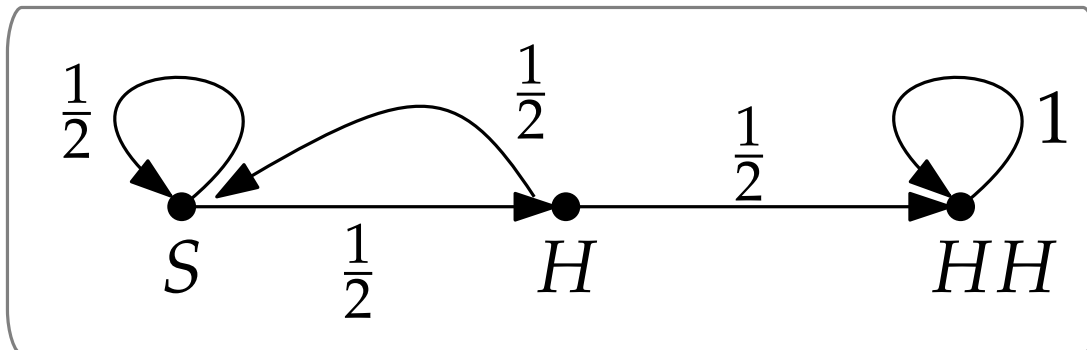
$$S \xrightarrow[2]{} H \xrightarrow[4]{} HH$$

Recurrence time

Is there a faster way?



$$S \xrightarrow[4]{} HT$$



$$S \xrightarrow[2]{} H \xrightarrow[4]{} HH$$

Since the Recurrence time is the time before a sequence *reappears again*, we need to check also that the subsequences *reappear again*.

Recurrence time

Consider the sequence $HTHTH$

- What is the probability that $HTHTH$ appears? $\frac{1}{32}$

Recurrence Time = 32

Expected Number of Flips = 42

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The answer to the ultimate question of life, the universe and everything!



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- What is the probability that $HTHTH$ appears? $\frac{1}{32}$

Recurrence Time = 32

Expected Number of Flips = 42

The answer to the ultimate question of life, the universe and everything!

No!

$$S \xrightarrow{2} H \xrightarrow{8} HT H \xrightarrow{32} HTHT H$$

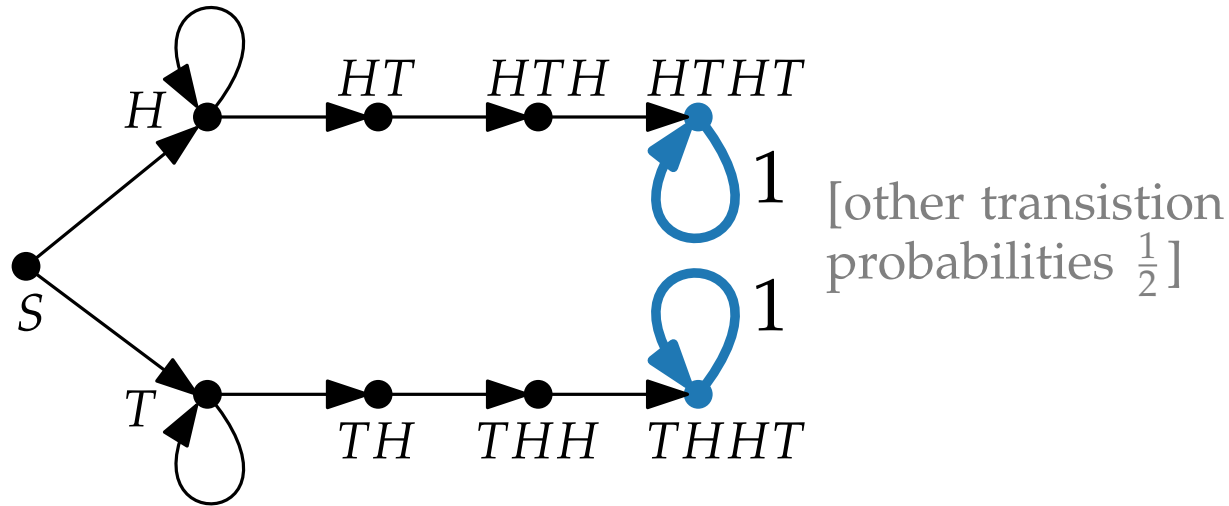


What's reached first?

Given two sequences of coin flips, which do you expect to reach first, e.g, between $HTHT$ and $THHT$?

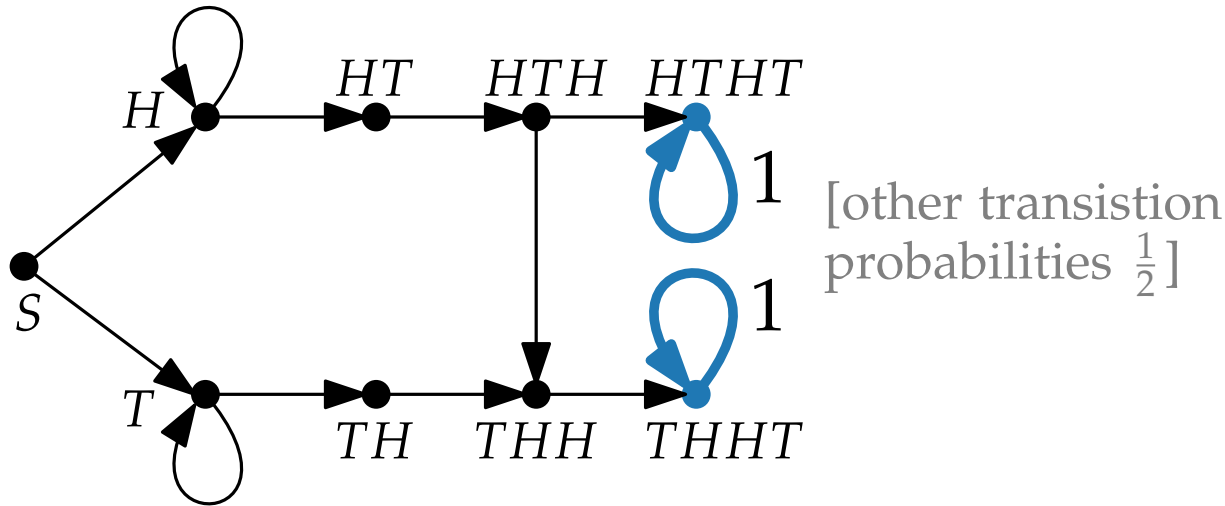
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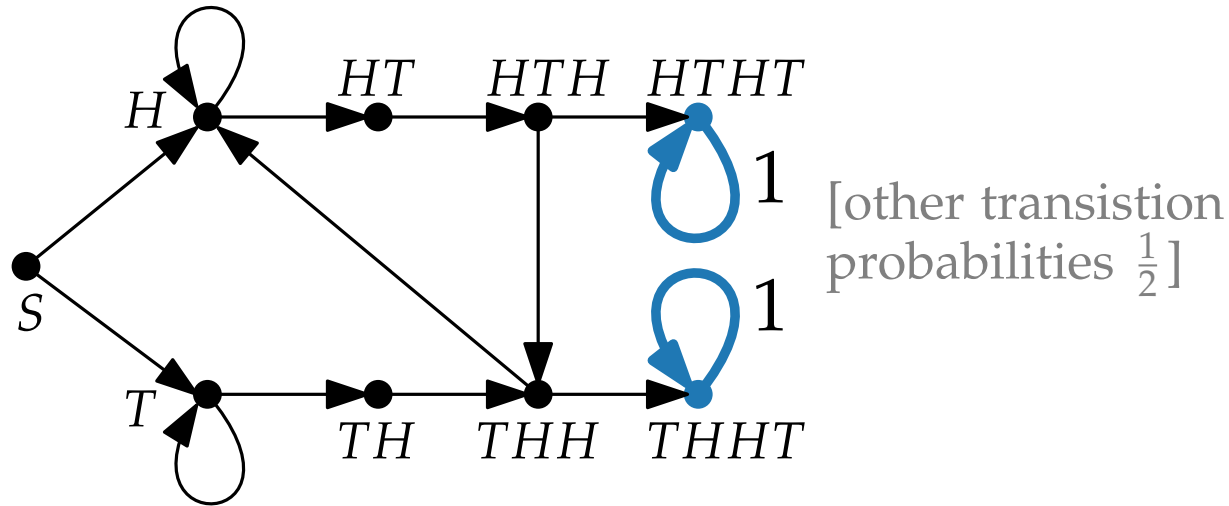
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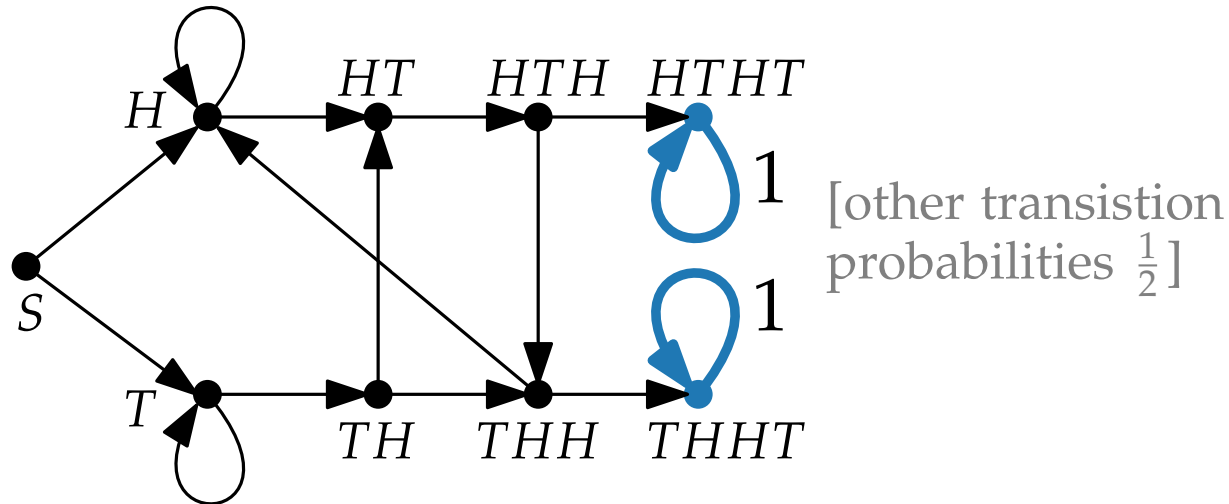
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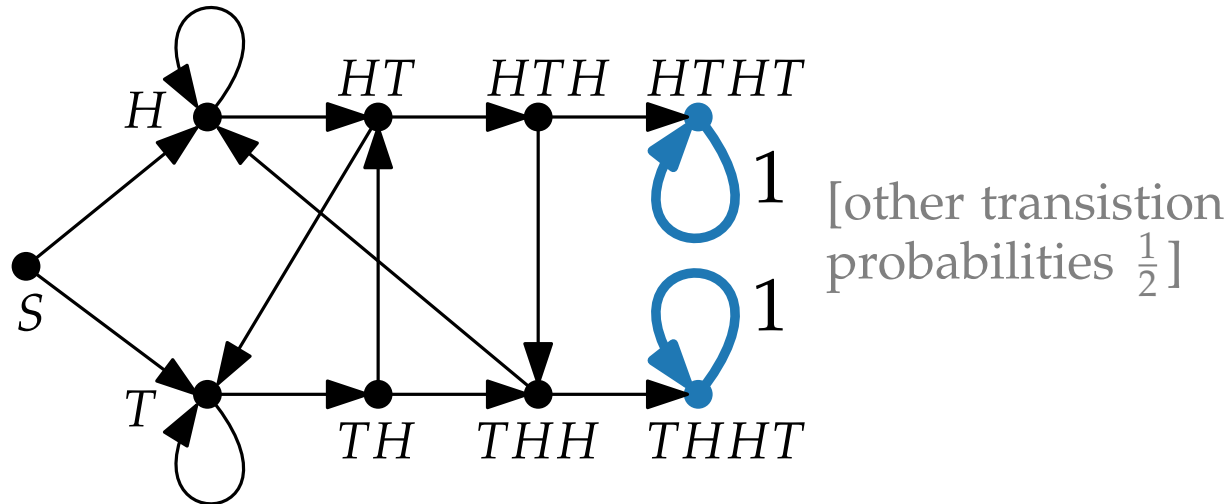
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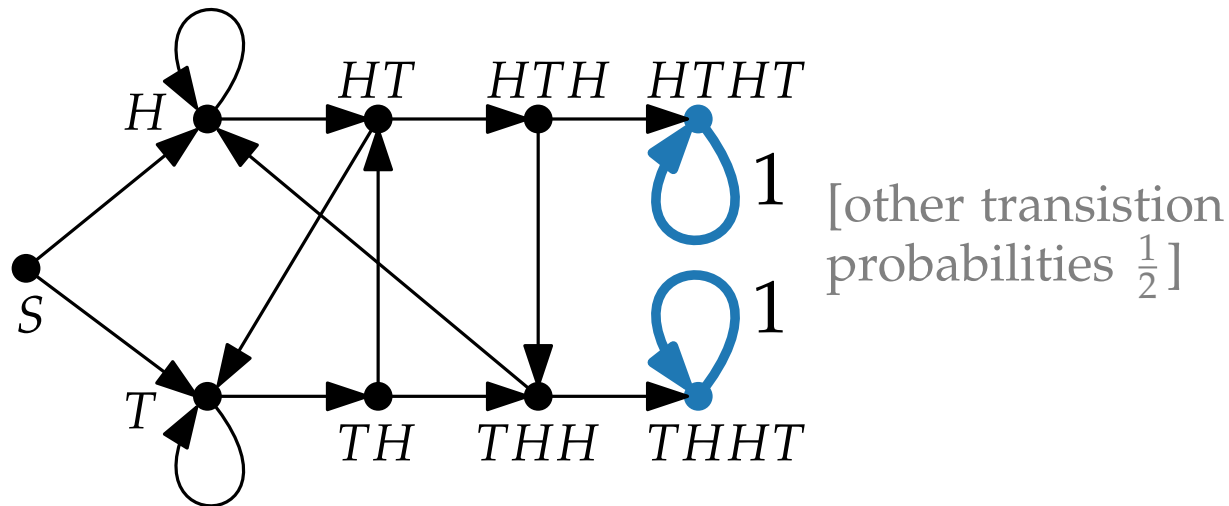
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What's reached first?

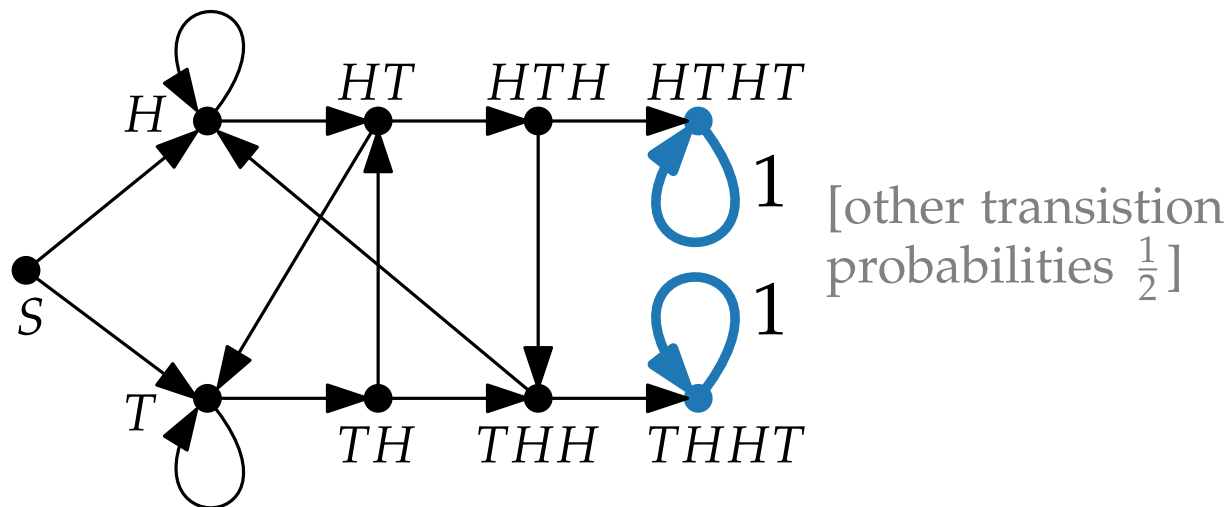
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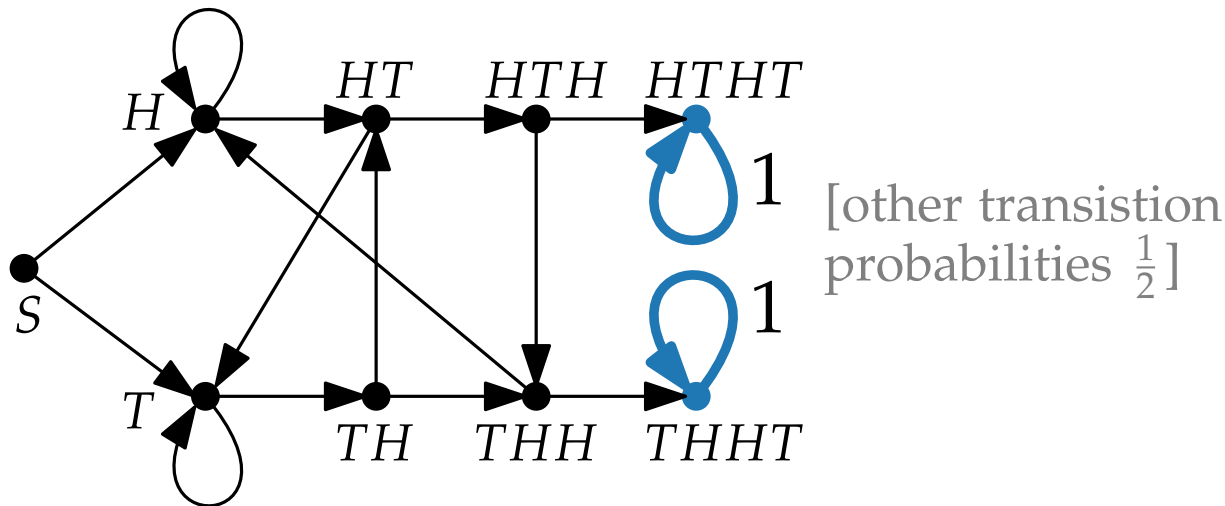
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...and so on

$$P_{THHT} = 0$$

$$P_{HTHT} = 1$$

Markov Chains

Given the present, the future is independent of the past.



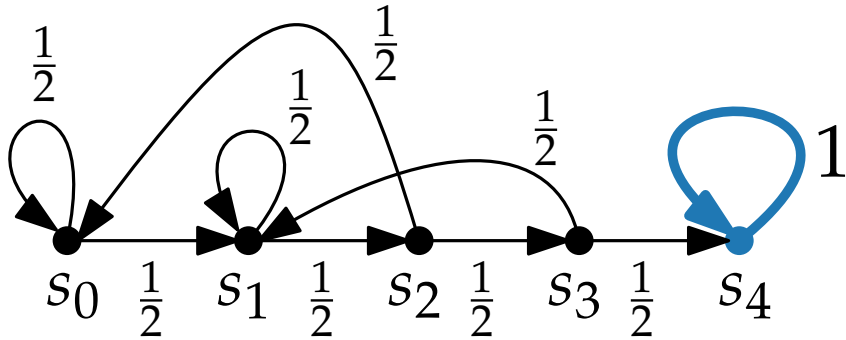
1856 –1922

Markov Chains

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1856 –1922



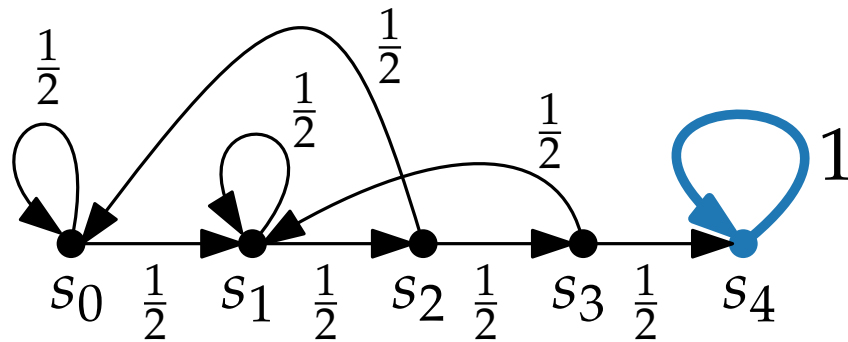
- *states:* s_0, s_1, s_2, s_3, s_4

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1856 – 1922



• states: s_0, s_1, s_2, s_3, s_4

Transition Matrix

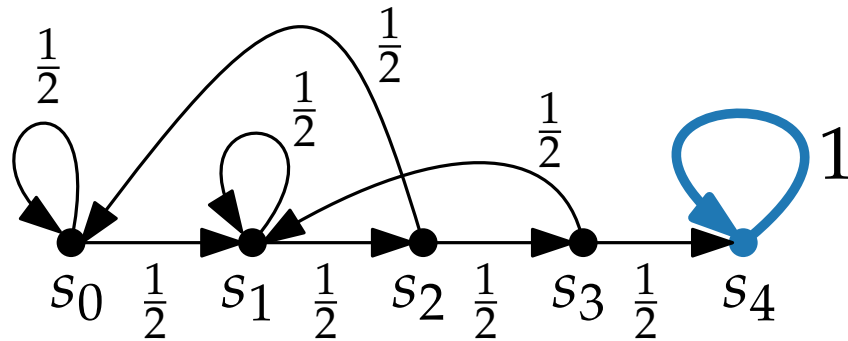
$$\begin{matrix} \text{from} \\ s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \end{matrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} \\ \\ \\ \\ \text{to} \end{matrix} \begin{matrix} s_0 \\ s_1 \\ s_2 \\ s_3 \\ s_4 \end{matrix} = P$$

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1856 – 1922



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to
 $s_0 \quad s_1 \quad s_2 \quad s_3 \quad s_4$

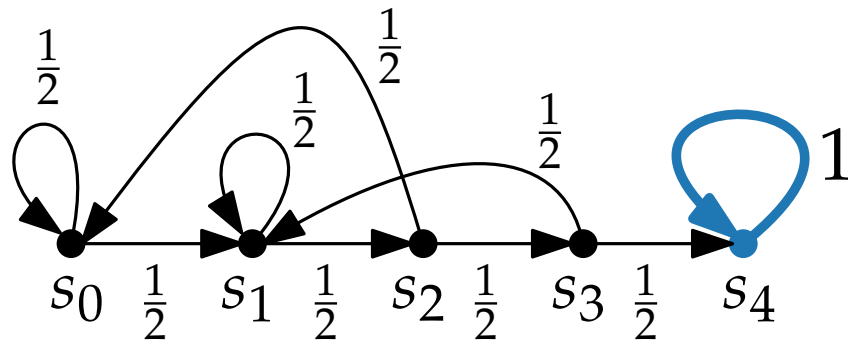
- *states*: s_0, s_1, s_2, s_3, s_4
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only visited **finitely often**

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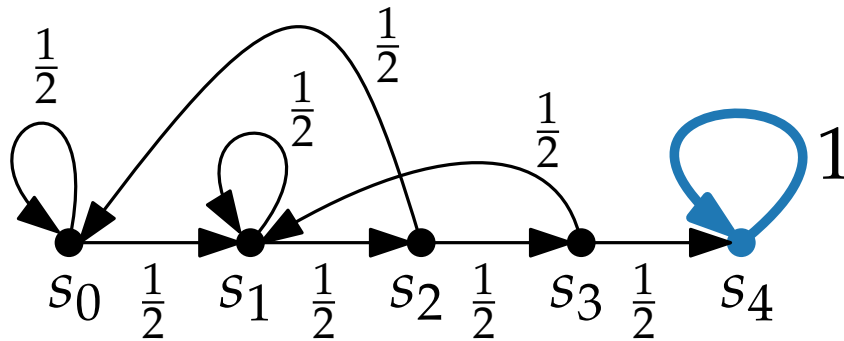
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1856 – 1922



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only visited **finitely often**
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- s_4 is also an *absorbing state* : it is inescapable.